AI with Rav

Learn AI the way you actually think.

DAY 2 of 30

MACHINE LEARNING

The 3 Families of Machine Learning

WHAT YOU'LL LEARN TODAY

- The 3 families of ML
- Supervised = learn with a teacher
- Unsupervised = find hidden groups
- Reinforcement = learn by reward

The 3 Families of Machine Learning

> In One Line: A machine learns in 3 ways — with a teacher, by exploring on its own, or by reward & punishment. Just like a child.

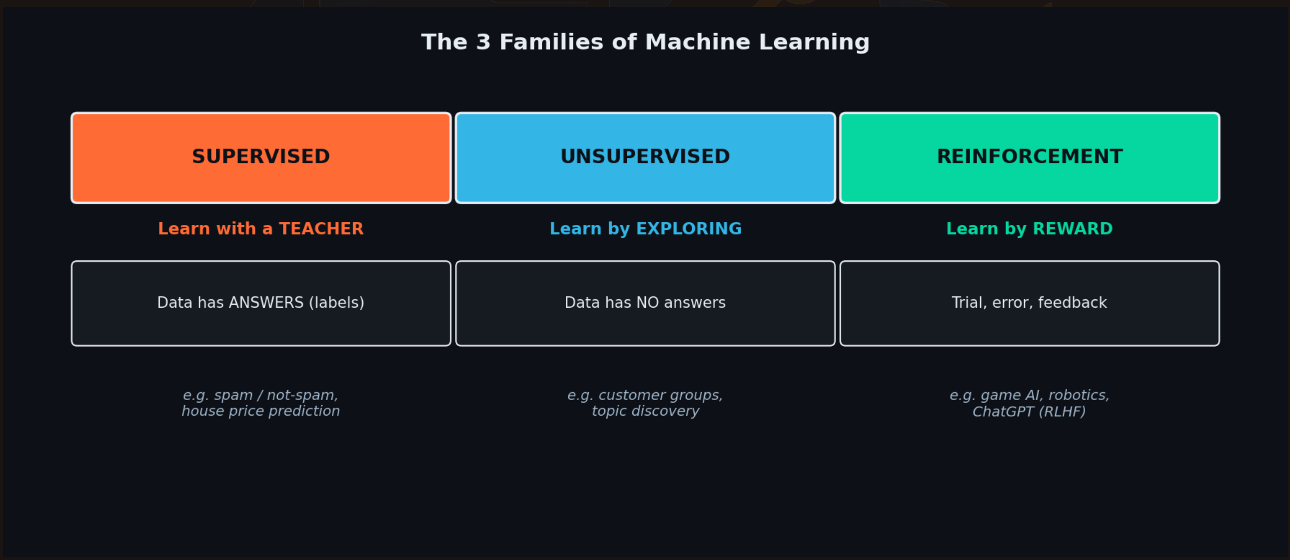
Start here: how a child learns

Think about how a child learns three different things:

- **Maths** → a teacher shows the answer: "2 + 2 = 4." The child learns from **labelled examples.**
- **Sorting toys** → nobody tells them the categories. They just notice "these are red, these are round" and **group by themselves.**
- **Cycling** → no teacher can hand it over. They try, fall, adjust — **learn from reward (staying up) and punishment (falling).**

Machines learn the exact same three ways. That's it — the whole map of ML.

The 3 families, side by side



Supervised (teacher), Unsupervised (explore), Reinforcement (reward). Every ML algorithm belongs to one of these.

1. Supervised Learning — learn with a teacher

You give the machine data **with the answers already attached** (called *labels*). It learns the pattern from answer → input, so it can predict answers for new data.

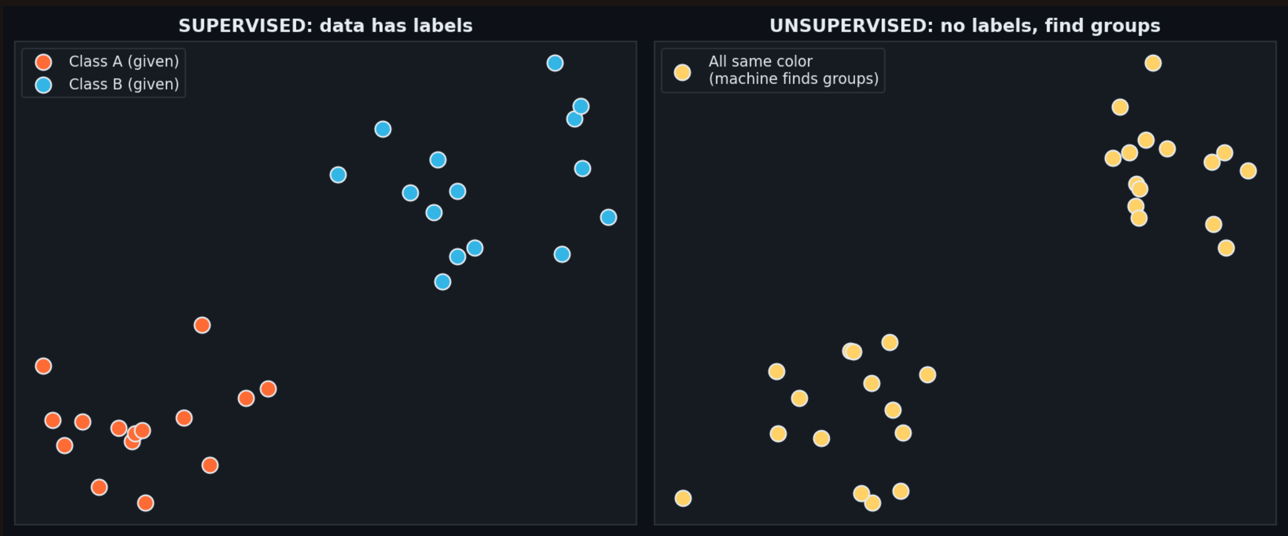
- Email marked *spam / not-spam* → learns to filter your inbox
- House features (size, location) + *actual price* → predicts price of a new house

- X-ray image + *doctor's diagnosis* → flags disease in a new scan

> Key sign: your data already has the right answers. You're teaching by example. Most business ML (fraud, churn, pricing) is supervised.

2. Unsupervised Learning — learn by exploring

Here the data has **NO answers**. The machine explores and finds hidden structure — groups, patterns — **by itself**, without anyone labelling anything.



Left: supervised — dots are pre-coloured (labelled). Right: unsupervised — all one color, the machine discovers the groups.

- A shop's customers → machine finds "budget buyers", "premium buyers", "festival-only buyers"
- News articles → auto-groups into sports, politics, tech (nobody tagged them)

> Key sign: no labels, you're asking "what natural groups or patterns are hiding in here?" Used for customer segments, recommendations, anomaly detection.

3. Reinforcement Learning — learn by reward

No teacher, no dataset. An **agent** tries actions, gets **reward** for good ones and **penalty** for bad ones, and slowly learns the best strategy — exactly like learning to cycle.

- Game AI → +1 for winning a point, learns to play better each round
- Robot arm → reward for picking the object correctly
- **ChatGPT** → RLHF: humans reward good answers, the model learns to prefer them

> Key sign: an agent taking actions in an environment, learning from trial-and-error feedback. This is how the "personality" of ChatGPT was trained.

Quick comparison

Family	Data has answers?	Learns from	Real example
Supervised	Yes (labels)	Examples with answers	Spam filter, price prediction
Unsupervised	No	Hidden structure	Customer groups, topic discovery
Reinforcement	No (gets reward)	Trial & error	Game AI, robots, ChatGPT (RLHF)

Recap — the 20-second version

- Machines learn 3 ways — like a child: teacher, exploring, reward.
- **Supervised** = data has answers → predict answers (most business ML).
- **Unsupervised** = no answers → find hidden groups/patterns.
- **Reinforcement** = agent learns from reward & punishment (game AI, ChatGPT).
- Ask one question to classify any ML problem: *"Does my data already have the answers?"*

> Next up — Video 3: The ML Workflow. How a real model is actually built, step by step — from raw data to a working prediction. See you Day 3.